

WELCOME

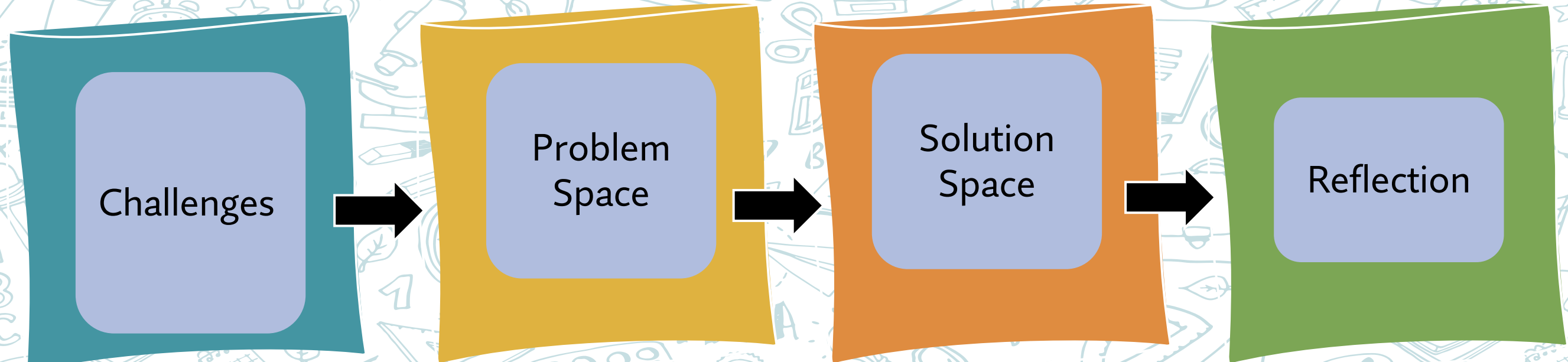
University Malaya Delegation

Mini Hackathon



UNIVERSITI
MALAYA

Mini Hackathon



Challenges

Produce a simple idea or prototype that addresses a shared social or digital issue

One group consists of 3 UM students + 2 ITB Students

Each group must choose and solve one theme

First 15 Minutes: Ideation

Next 15 Minutes: Brainstorming

Next 10 Minutes: Converge the idea

Next hours: Build prototype

Present the results

Themes to solved

Each group (3 UM +2 ITB) need choose one problems :

Smart City & Mobility

Indonesia: Big cities like Jakarta and Surabaya struggle with severe traffic congestion, unpredictable public transport, and limited real-time travel information. People waste hours commuting every day.

Malaysia: In Kuala Lumpur and Penang, traffic jams and parking shortages are daily frustrations. Travelers often lack accurate updates about road conditions or transport schedules.

Problem: How can technology help citizens move more efficiently, reduce congestion, and improve safety in urban areas?

Theme to solved



Each group (3 UM +2 ITB) need choose one problems :

Health & Wellbeing

Indonesia: Unequal access to health information and services makes it hard for people in smaller cities or rural areas to stay informed about disease prevention and healthy lifestyles.

Malaysia: Lifestyle-related illnesses such as obesity and diabetes are increasing, while awareness about nutrition and fitness remains low.

Problem: How can digital solutions encourage healthier living and make health information more accessible to everyone ?

Theme to solved



Each group (3 UM +2 ITB) need choose one problems :

Education & Learning Tools

Indonesia: Many students, especially outside major cities, lack access to engaging and up-to-date learning materials. Teachers also need better tools to personalize lessons.

Malaysia: Digital learning is growing, but students often struggle to stay motivated and find content that matches their learning style.

Problem: How can technology make learning more interactive, inclusive, and personalized for students in both countries?

Theme to solved



Each group (3 UM +2 ITB) need choose one problems :

Tourism & Culture Exchange

Indonesia: Many local destinations remain underpromoted, and travelers often miss out on cultural experiences beyond the main tourist spots.

Malaysia: Cultural diversity is rich, but cross-border understanding and collaboration with neighboring countries are still limited.

Problem: How can digital innovation strengthen cultural exchange and make travel experiences more meaningful between Indonesia and Malaysia?

Theme to solved



Each group (3 UM +2 ITB) need choose one problems :

Sustainability & Environment

Indonesia: Air pollution, waste management, and deforestation are major environmental challenges, yet public awareness and community action remain low.

Malaysia: Flooding and environmental degradation continue to affect communities, with limited access to timely information and education on sustainability.

Problem: How can technology help people understand environmental issues better and take action toward a more sustainable future?

Looking for Ideas

Looking for Ideas

Searching for **ideas** in design is an exploration of various ways and options to address challenges with multiple solutions. Play around!

The essence of Ideation in Design Thinking

How can we feel **comfortable**
to generate ideas

As many as possible!

As quickly as possible!

Even if it looks **chaotic** at first!

The Art of Ideation

Freedom is essential in the early phase of ideation — it's okay **to play around, be wild, and even a little crazy.**

“Be playful”

“I believe the opposite of play, is boredom”

-Brendan Boyle, IDEOU

“The best way to have a good idea is to have lots of ideas”

-Linus Pauling, two times Nobel Prize Winner

Several creative methods for generating ideas **[ideation]** in design:

Brainstorming

Generating as many ideas as possible in a short time, while postponing judgment of their quality.

Mash-Up

Combining two things to see whether the functions or benefits of both can become a new solution.

Example: Hospital + Hotel = Hotel-style Hospital Room
Car + Airplane = Flying Car

Put-on other people shoes

Role-playing or creating a storyboard/sketches of possible imagined use-case scenarios.

E-storming

Using online media (email, social media) to crowdsource opinions about a particular challenge.

...
(and many more)

Brainstroming

A way to solve a problem by **asking group members** to share creative ideas **freely** and **spontaneously**



*“using the **brain** to **storm** a problem”*

Metode **Pemecahan Masalah** secara **Kreatif (Creative Problem Solving - CPS)** yang paling populer, dikenalkan oleh **Alex F. Osborn**, advertising executive, US, pada 1940-an.

Tools:

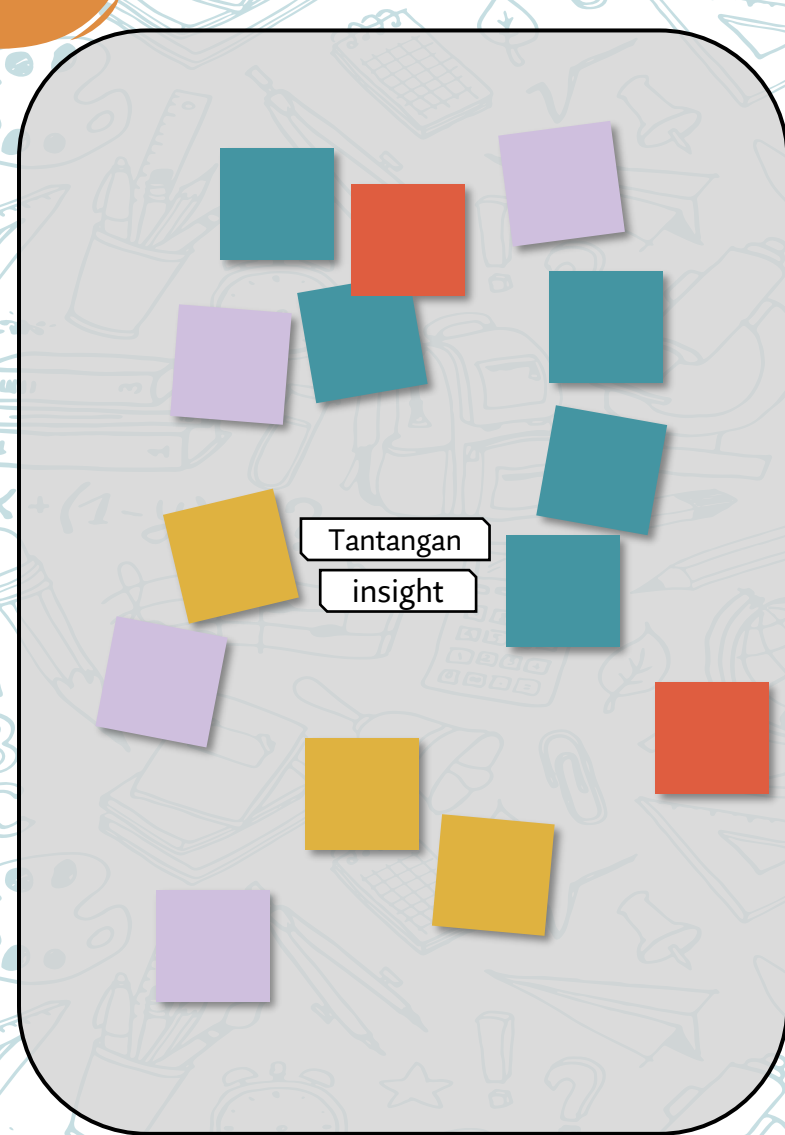
Small adhesive note sheets – sticky notes (e.g., Post-it)

Large board (with a canvas for support)

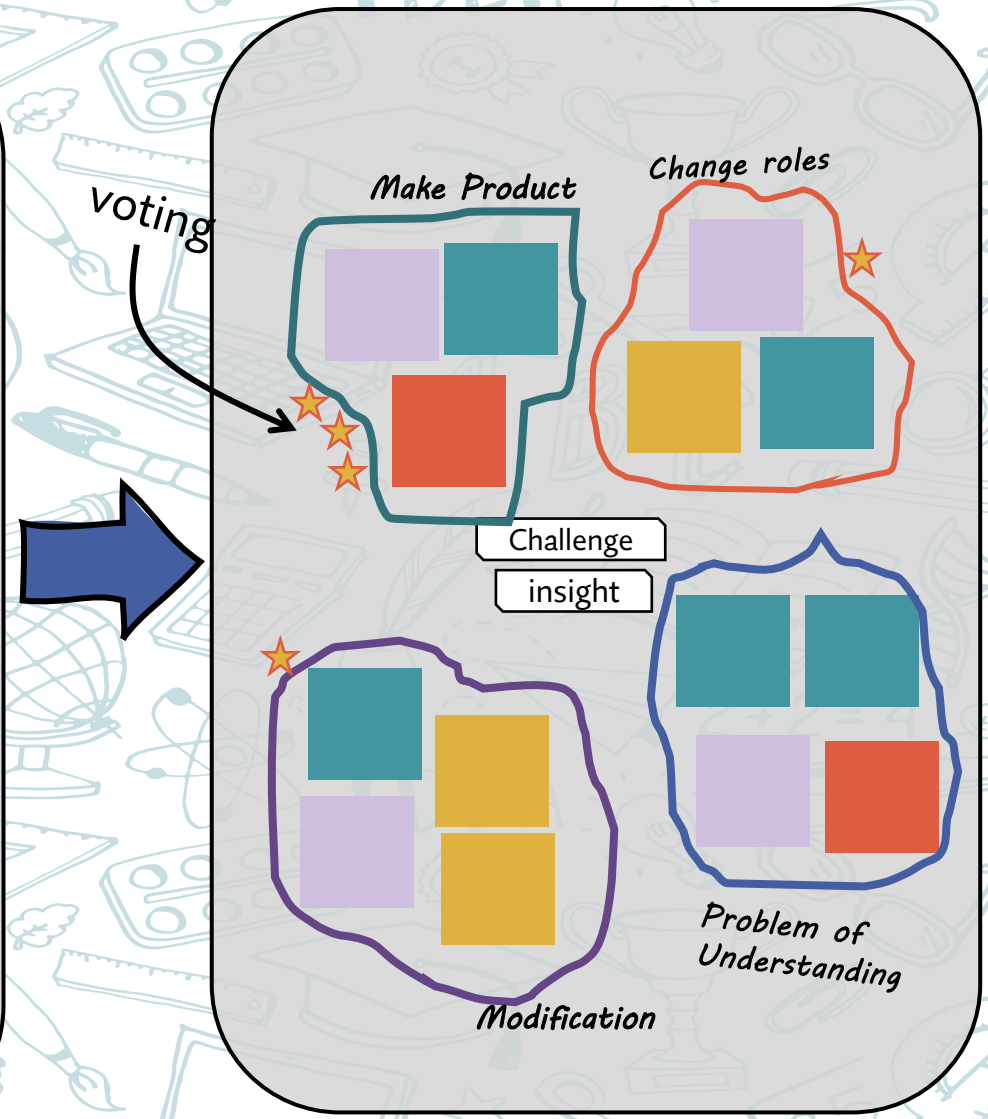
Writing tools

Timer

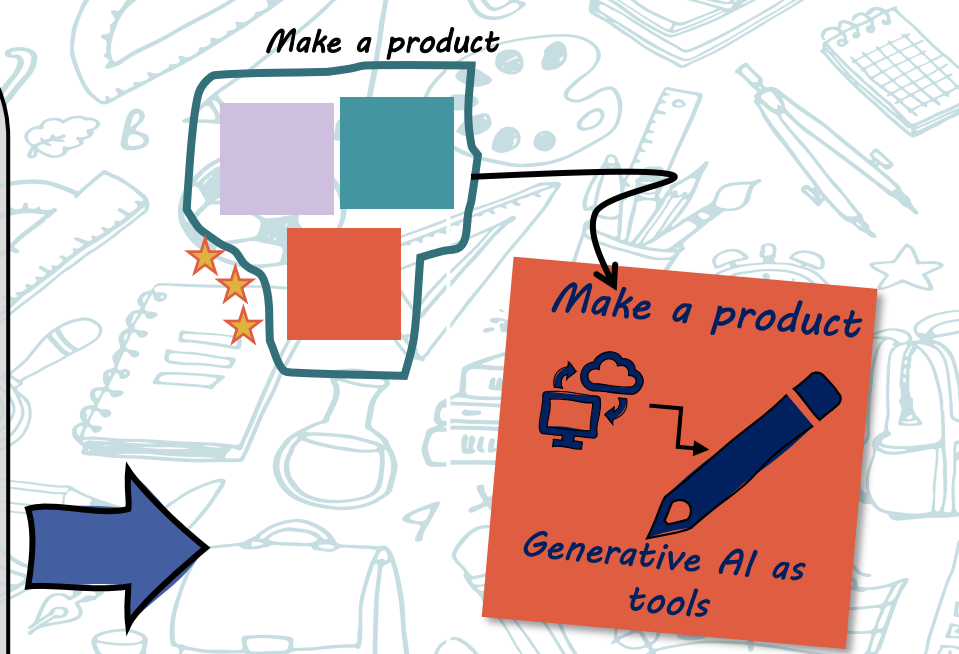
Brainstorming Session



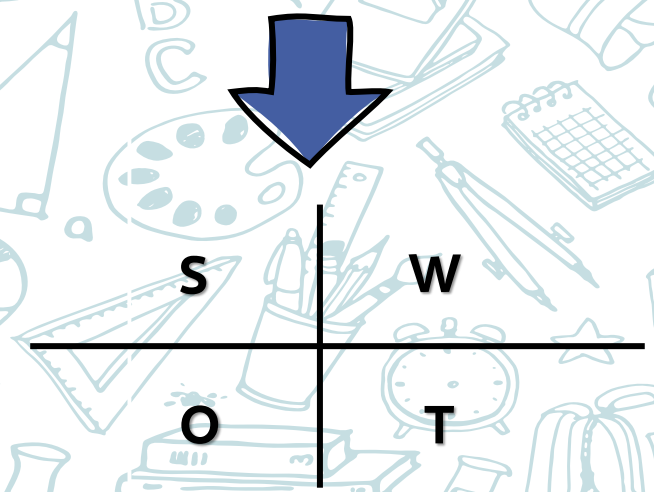
Brainstorm!
Suggestion: 2 x 15 minutes



Organize Group
Vote



Choose/summarize the best idea



Do a simple analysis

How to Brainstorm Effectively

Rules of Brainstorming - IDEOU



1. Defer Judgment/Criticism

The creative space at the beginning **must be free of criticism**. Let ideas flow first, and let ideas emerge from other ideas.



2. Welcome Wild Ideas

Support the most **out-of-the-box** ideas. There is often a thin line between **crazy ideas** and **brilliant ones**.



3. Build on the Ideas of Others

Try using "**and...**" instead of "**but...**" when commenting on ideas. This encourages a **positive and conducive** atmosphere for new ideas to emerge.



4. One Conversation at a Time

This is harder than it sounds, especially if the group is full of creative people. But it is absolutely necessary.



5. Stay Focused on the Topic

Try to **stay focused** on the challenge topic and the **insights** that have been selected.



6. Be Visual

Don't just use text; create **sketches** with colorful markers on sticky notes. Post them in **clear positions** on the canvas.



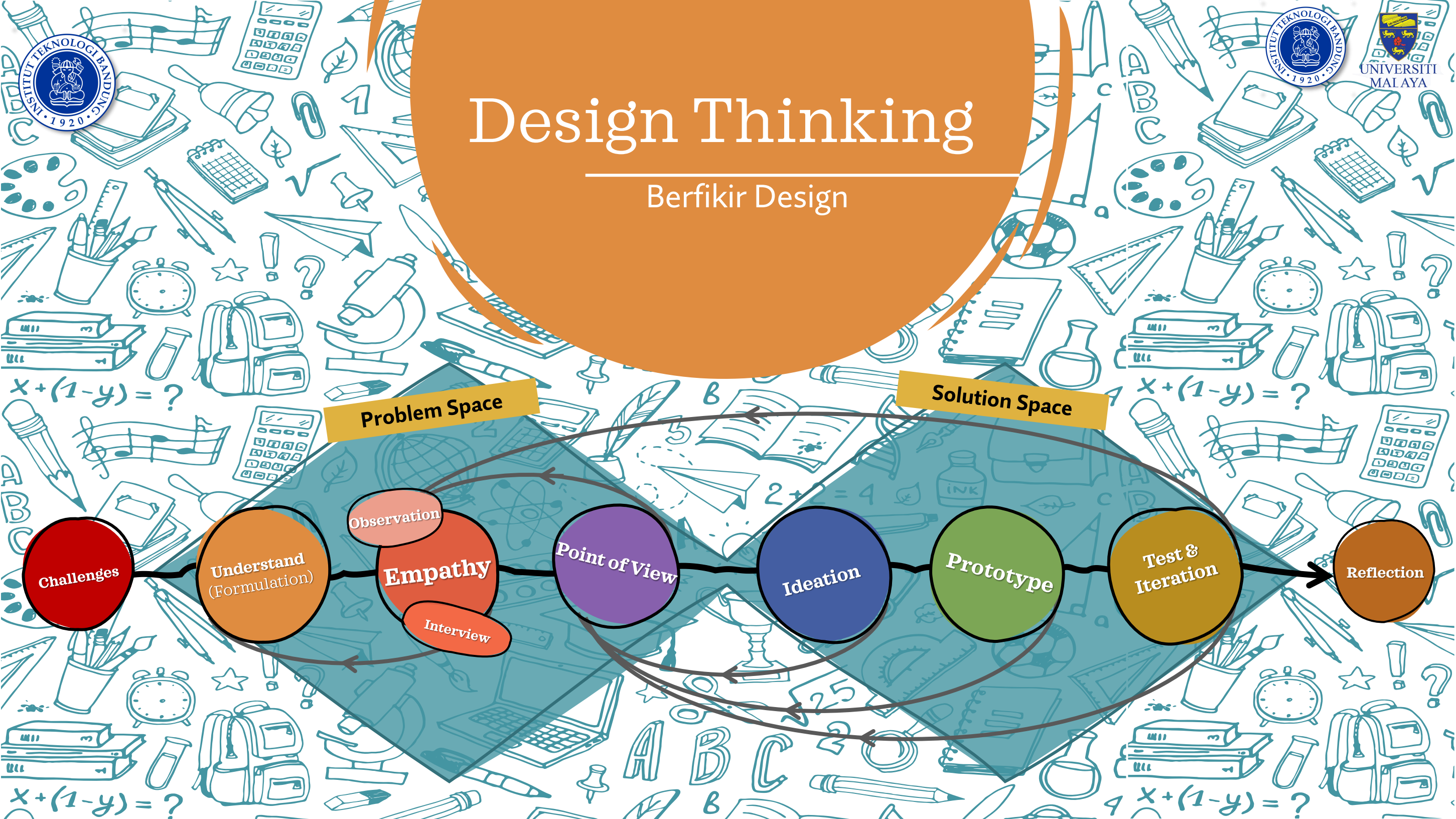
7. Go for Quantity

The goal is to **generate ideas as quickly** as possible. For a 2 x 10 minute session, try to create more than 50 ideas



Design Thinking

Berfikir Design



How to Brainstorm Effectively

[Example] Technical Procedure for Rapid Brainstorming

Set the Conditions

- Choose a **board/canvas** of sufficient size. Ensure the canvas is **mounted properly** and is **sturdy**.
- Ensure all participants have enough **sticky notes** and **markers** that are in good working order.
- Write down the **challenges** and **key insights** that can help the group stay **focused** on the canvas.
- **One person explains** the **background** and ensures the procedure is clear (the importance of a timer) and explains the rules to all participants

Execute! session 1, 15 minutes

- Take turns, **but one after another**, participants *express their ideas*, write them on **sticky notes**, and stick them on the canvas.
- **Ideas** that arise **from listening** to *other participants' ideas* are highly important.
- Starting with **wild (absurd) ideas** is normal.

STOP - Take a 2-minute break.

Execute again! session 2, 15 minutes

Converge! 10 minutes

- Find ideas **that are similar**, related, or share the same theme; **group them together** in one spot, circle them with a marker, and add a label.
- **Conduct a voting** session (by marking with stars) to determine which circle is the best.
- Take **the best sticky note** from the best circle (or create a new note summarizing them all).
- Do a **brief analysis** of the idea's potential (such as a **simple SWOT analysis**).

Prototypes for Testing

The next Testing / Iteration step → ask potential users, request feedback

But in this step, a **testable prototype** is required!

Critical Function Prototype (CFP)

What is tested?

Focus on **part-by-part functions**, generally so it can be directly **tested on users** / against certain procedures

Those focusing on user experience → **Critical Experience Prototype (CEP)**

Example:

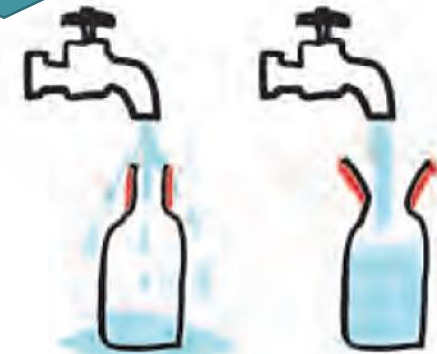
How Might **We** better fulfill the water needs for **hikers** during a long mountain climb??

SOLUTION: A water bottle that continues to function well even when hikers are **tired** and **sweaty**

4 Critical Function



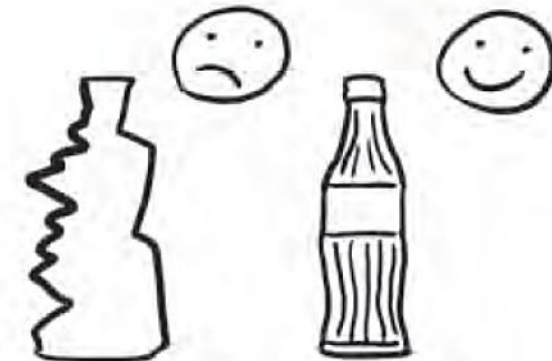
Critical function F1 "GRIP"



Critical Experience E3 "FILL"

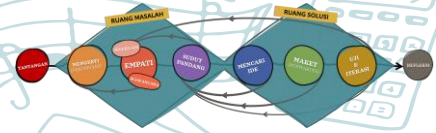


Critical function F2 "OPEN"



Critical Experience E4 "DESIGN"

Kombinasi Solusi



Bottle Prototype for Hikers

Holding it when hands are sweaty



Use rubber rings/grips

Opening / Access to drink / refill



Flexible bottle neck

Easy to carry



With a belt



like a Coca-Cola bottle



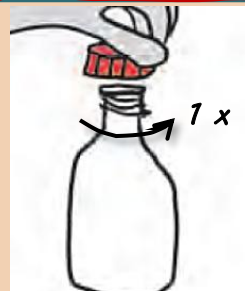
Large cap!



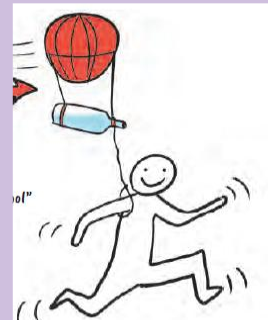
Backpack (injected?)



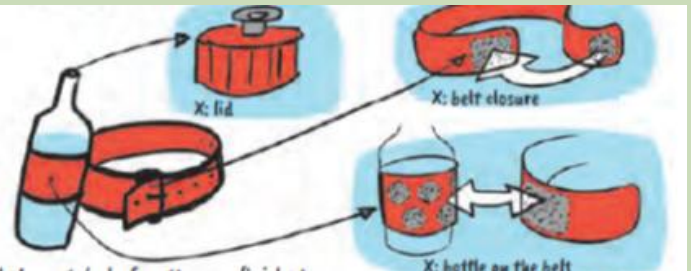
With a handle



Cap that only twists once



Balloon



Looking for Ideas

Vibe Coding



Tools Needed

Node JS

<https://nodejs.org/en/download>

Gemini CLI

Run as Administrator PowerShell

```
Set-ExecutionPolicy RemoteSigned -Scope CurrentUser
```

Then

```
npm install -g @google/gemini-cli
```


Thank You



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